Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/20/2024
PI Name:	Jacob, Naduparambil K Ph.D.		
Project Title:	Predictive Biomarkers for Space Radiation Induced Cancer and Cardiovascular Injury Risk Assessment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	614-685-4246
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City:	Columbus	**State:**	OH
Zip Code:	43210	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	07/02/2018	**End Date:**	12/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per NSSC information (Ed., 10/29/23) NOTE: End date changed to 7/1/2023 per NSSC information (Ed., 4/3/23) NOTE: End date changed to 7/1/2022 per NSSC information (Ed., 6/22/21) NOTE: End date changed to 7/1/2021 per NSSC information (Ed., 9/25/20)		
Key Personnel Changes/Previous PI:	April 2021 report: Dr. Peter Lee is no longer CoInvestigator on the project.		
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K1691		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	To reduce the uncertainty in estimates of cancer and cardiovascular risks from space radiation, we will evaluate changes in molecular biomarkers in rodents and rabbits exposed to ions relevant to exposures of astronauts in the space environment. Serum, heart, liver, and lung tissues collected from exposed animals and matching controls (as available) will be used for biomarker discovery following systems-biology approaches. The study will use modern analytic technologies and rigorous statistics for assessing changes in expression of microRNAs (miRNAs) associated with clinical endpoints for mechanistic understanding of disease initiation and progression. Analysis of samples from patients receiving radiation therapy and organ targeted and organ protected low-linear energy transfer (LET) irradiation model studies have shown changes in circulating miRNAs originating from organ systems as a function of dose exposed and time. Circulating miRNAs collected from mice exposed to low doses of neutrons will be compared with gamma rays and sham controls of changes in cancer endpoints. Cellular and molecular mechanisms involved in space radiation-induced cardiovascular disease, and cancer will be studied, which will contribute to risk assessment and developing effective countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The goal is to develop blood test for early detection of delayed cardiovascular complications and cancers resulting from high-LET radiation exposure to astronauts during long duration space travel.
Task Progress:	To reduce the uncertainty in estimates of cancer and cardiovascular risks from space radiation, we evaluated changes in molecular biomarkers in rodents and rabbits exposed to ions relevant to exposures of astronauts in the space environment. Serum, heart, liver, and lung tissues collected from exposed animals (two strains of mice, rabbit, and rats) and matching controls were used for biomarker discovery following systems biology approaches. The study used modern analytic technologies and rigorous statistics for assessing changes in expression of microRNAs (miRNAs), long noncoding RNAs (lncRNAs). In the aim 1, archived cardiac specimens from rabbits and rats previously exposed to 0.5 Gy Proton or Oxygen ions with respective sham controls were used for discovery and validation of space radiation-induced changes in miRNAs. In the aim 2, miRNAs and lncRNAs in lung, liver, and serum collected two strains of mice exposed to low dose rate neutron were compared with sham controls of changes.
Bibliography Type:	Description: (Last Updated: 10/14/2024)